GRADE 7 SCIENCE · ENERGY: HEAT

How Heat Travels

Conduction, convection and radiation

What you will learn

After this lesson, you can:

- tell the difference between heat and temperature;
- describe conduction, convection and radiation;
- give an example of each one from daily life;
- explain why some materials are used to keep things hot or cold.

1. Heat and temperature

Hold a glass of iced water and your hand feels cold. Hold a cup of hot chocolate and your hand feels warm. In both cases, energy is moving.

Temperature tells us how hot or cold something is. We measure it with a thermometer, in degrees Celsius (°C). **Heat** is energy that moves from one object to another because their temperatures are different. Heat always flows from the hotter object to the cooler object. It keeps flowing until both objects have the same temperature.

So when you hold a glass of iced water, coldness does not move into your hand. Heat moves out of your warm hand and into the cold glass. That is why your hand feels cold.

Heat can travel in three ways: conduction, convection and radiation.

2. Conduction: heat passed on by touch

Aling Rosa leaves a metal ladle in a pot of hot sinigang. After a few minutes, the handle is too hot to hold, even though the handle was never in the soup. Heat travelled along the ladle.

Conduction is the transfer of heat through direct contact. All matter is made of tiny particles that are always moving. When one end of the ladle is heated, its particles vibrate faster. They bump into the particles next to them and pass the energy along. The particles stay in their places. Only the energy moves, from particle to particle.

Conduction works best in solids, because the particles of a solid are packed closely together.

Materials that let heat pass through them easily are called **conductors**. Metals such as copper, aluminum and iron are good conductors. This is why pots and pans are made of metal.

Materials that do not let heat pass through them easily are called **insulators**. Wood, plastic, rubber, cloth and air are insulators. This is why a pan has a wooden or plastic handle, and why we use a cloth pot holder to lift a hot kaldero.

3. Convection: heat carried by moving liquids and gases

Liquids and gases are called **fluids** because they can flow. In fluids, heat travels mainly by convection. **Convection** is the transfer of heat by the movement of the fluid itself.

Watch a pot of water on a stove. The water at the bottom is heated first. It expands, becomes less dense and rises. The cooler water at the top is denser, so it sinks to the bottom, where it is heated in turn. This circular movement is called a **convection current**. It continues until all the water is hot.

Convection also happens in air. Near the sea, it causes breezes:

- During the day, the land heats up faster than the sea. Warm air above the land rises, and cooler air from the sea moves in to take its place. This is a **sea breeze**.
- At night, the land cools faster than the sea. Now the air above the sea is warmer and rises, and cooler air from the land moves out toward the sea. This is a **land breeze**.

Fishers in sailboats have long used these breezes. The land breeze helps them sail out before sunrise, and the sea breeze helps them come home in the afternoon.

4. Radiation: heat that needs no matter

The Sun is about 150 million kilometers from Earth, and the space in between is almost empty. There are no particles to touch and no fluid to flow. Yet the Sun's heat reaches us. It travels by radiation.

Radiation is the transfer of heat by electromagnetic waves. Unlike conduction and convection, it does not need matter. It is the only way heat can travel through empty space. You feel radiation when you sit near a charcoal fire or stand under the midday sun. Farmers use it when they spread palay on the pavement to dry.

Surfaces do not all take in radiation equally. Dark, dull surfaces absorb more heat. Light-colored and shiny surfaces reflect more of it. This is why a black shirt feels hotter under the sun than a white shirt, and why many roofs are painted white or silver to keep houses cooler.

5. All three at once

In daily life, the three ways often happen together. Think of boiling water in a metal pot over a charcoal stove:

- **Radiation:** the glowing charcoal sends heat to the bottom of the pot, and to your face when you lean close.
- **Conduction:** heat passes through the metal of the pot to the water that touches it, and along the handle.
- **Convection:** the hot water rises and the cooler water sinks, which spreads the heat through the pot.

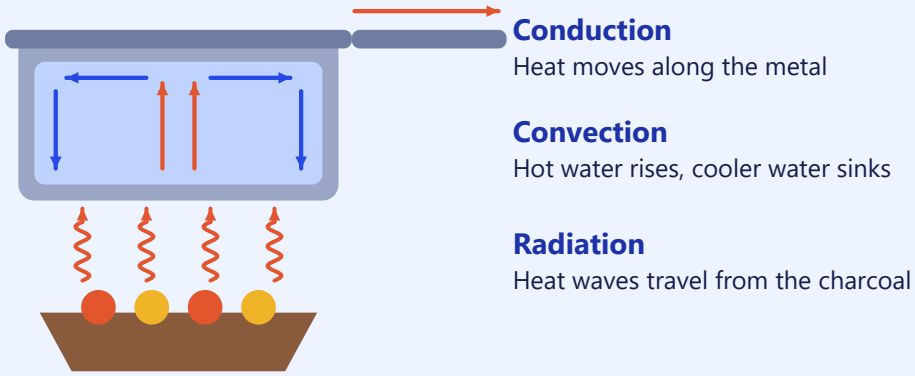


Figure 1. Boiling water over charcoal uses all three ways of heat transfer.

The three ways at a glance

Way	How the heat moves	Where it happens	Example
Conduction	From particle to particle, by direct contact	Mostly in solids	A metal ladle gets hot in soup
Convection	Carried by a fluid that moves	In liquids and gases	Boiling water; a sea breeze
Radiation	By electromagnetic waves	Even through empty space	Heat from the Sun

6. Keeping things hot or cold

When we know how heat travels, we can control it.

A vacuum flask, or thermos, keeps coffee hot for hours. It has two walls with almost no air between them, so very little heat escapes by conduction or convection. Its shiny inner surface reflects radiation back into the drink.

A styrofoam ice box keeps fish cold on the way to the market. Styrofoam is full of tiny pockets of trapped air. Air is a poor conductor, so heat from outside enters the box slowly.

A house with windows on opposite sides, or with vents near the roof, stays cooler. Warm air rises and leaves, and cooler air comes in to replace it. That is convection at work.

Remember

- Heat is energy that flows from a hotter object to a cooler object.
- Conduction is heat transfer by direct contact. It works best in solids, and metals are good conductors.
- Convection is heat transfer by a moving liquid or gas. Warm fluid rises and cool fluid sinks, which forms a convection current.
- Radiation is heat transfer by electromagnetic waves. It can travel through empty space.
- Insulators such as wood, plastic, cloth and trapped air slow down the transfer of heat.